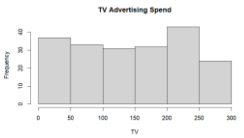
Task 1:

There are 200 observations and 4 variables in this data set: TV, Radio, Newspaper & Sales The first 3 variables are the advertising expenses (in thousands of dollars). The variable Sales is the sales of product (in thousands of units).

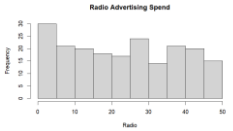
Table (1) Summary statistics for TV, Radio, Newspaper and Sales

Variable	Mean	SD	Min	Q1	Median	Q3	Max
TV	147.04	85.85	0.70	74.38	149.75	218.82	296.40
Radio	23.26	14.85	0.00	9.97	22.90	36.52	49.60
Newspaper	30.55	21.78	0.30	12.75	25.75	45.10	114.00
Sales	15.13	5.28	1.60	11.00	16.00	19.05	27.00

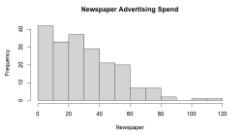
Table 1 shows the mean, standard deviation, minimum, first quartile, median, third quartile & maximum for each variable in the dataset



The range of TV values is 0.7 to 296.4 with a mean of about 147.04. This indicates the TV has the highest spending in advertising and that TV advertisements are the most variable.



The Radio values range from 0 up to 49.6 with a mean close to 23.26. This indicates Radio has the lower advertising budget, but the values are more clustered.



The Newspaper values range from 0.3 up to 114, with a mean close to 30.55. The histogram is right skewed in which most values have the lower of the spending and the few observations have the highest value.



The Sales are 1.6 to 27 with an average of about 15.13. The histogram indicates that most of the activity is in the middle band approximately 10 to 20.

The plots above are, therefore, only a very modest start in summarising the data on advertising budgets and sales in the data set.

Task 2:

The correlation between the variables is presented by a correlation matrix.

```
> cor(advertising.csv)
                 TV      Radio  Newspaper     Sales
TV        1.00000000 0.05480866 0.05664787 0.9012079
Radio     0.05480866 1.00000000 0.35410375 0.3496311
Newspaper 0.05664787 0.35410375 1.00000000 0.1579600
Sales     0.90120791 0.34963110 0.15796003 1.0000000
>
```

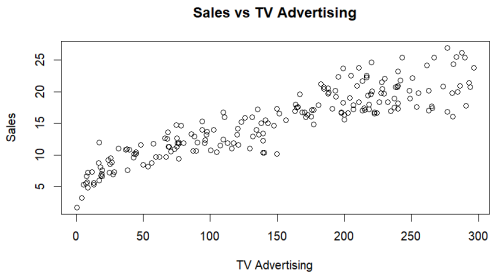
The correlation between TV and Sales =0.901 So, the relationship between TV and Sales is strong positive, that means as the spending on sales increases, the sales increase.

The correlation between Radio and Sales is 0.350. Therefore, there is a moderate positive relationship between Radio and Sales i.e. as the spending on sales increases, the sales increase, but weak compared to TV.

The correlation between Newspaper and Sales is 0.158. Therefore, there is a weak positive relationship between Newspaper and Sales i.e. as the spending on sales increases, the sales increase, but weak compared to Radio. The correlation between the advertising channels themselves is low, with TV having very weak relationships with both Radio and Newspaper and Radio and Newspaper having a moderate correlation of 0.354.

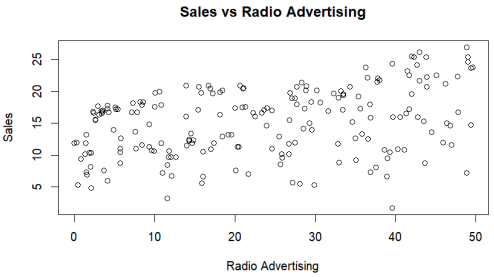
Consequently, there is more correlation between TV and Sales than between the other two advertising media.

Figure 5. Sales vs TV Advertising



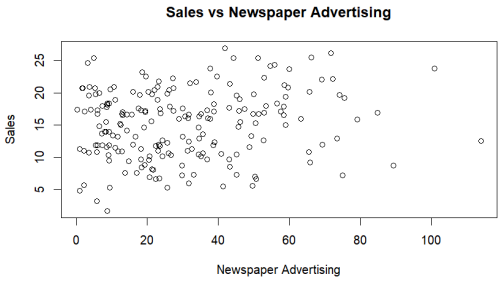
The scatterplot of sales and TV advertising shows that there is a positive linear relationship, as sales increases TV advertisement increases. The points are moving upward. This is in consistency with high correlation value of about 0.90 in correlation matrix.

Figure 6. Sales vs Radio Advertising



The scatterplot of sales and radio advertisement shows that there is an intermediate relationship. as sales increase the radio advertisement tends to increase. However, the points are scattered distantly from each other unlike the TV. This is in consistency with intermediate correlation value of about 0.35.

Figure 7. Sales vs Newspaper Advertising



The scatterplot of sales and newspaper shows that there is a weak relationship. The points do not tend to rise rapidly. This is in consistency with weak correlation of about 0.16 that newspaper advertising is only weakly associated with sales.

Conclusion:

It seems that sales and TV advertisement have the most association, sales and radio advertisement has an intermediate association, sales and newspaper advertisement has the least association.

Table (2) Pearson correlation tests between predictor variables

Variable Pair	Correlation Coefficient	p-value	Interpretation
TV and Radio	0.0548	0.4408	No statistically significant association
TV and Newspaper	0.0566	0.4256	No statistically significant association
Radio and Newspaper	0.3541	2.689 × 10^-7	Statistically significant moderate positive association

The Pearson Correlation Tests for the Predictor Variables (Table 2) shows that TV is not associated in a statistically significant way with Radio nor Newspaper. Radio and Newspaper have a moderate positive relationship with each other that is statistically significant. Thus, TV is independent of the other two predictors. Radio and Newspaper are moderately related to each other.

Task3 (Simple Linear Regression Model)

Model specification:

Sales = β0 + β1 × TV
From the R output:

Intercept = 6.9748

TV coefficient = 0.05546

Estimated regression equation

```
Console   Terminal   Background Jobs
R 4.5.2 · ~/
lm(formula = Sales ~ TV, data = advertising.csv)

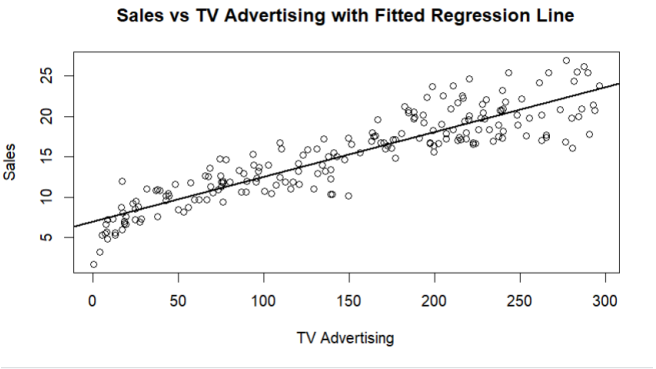
Residuals:
     Min       1Q   Median       3Q      Max
-6.4438 -1.4857   0.0218   1.5042   5.6932

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 6.974821   0.322553   21.62   <2e-16 ***
TV          0.055465   0.001896   29.26   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.296 on 198 degrees of freedom
Multiple R-squared:  0.8122,    Adjusted R-squared:  0.8112
F-statistic: 856.2 on 1 and 198 DF,  p-value: < 2.2e-16
```

Sales = 6.9748 + 0.05546 × TV

Figure 8. Sales vs TV Advertising with fitted regression line



The scatterplot with fitted regression line illustrates a positive linear relationship between TV advertising and sales. A positive slope of the fitted line suggests TV advertising is a positive predictor of sales, which accords with seeing that TV advertising is a strong positive predictor of sales.

Interpretation:

The intercept value of 6.9748 tells us the predicted sales when we spend zero dollars on TV advertising. Even if no money is spent on TV advertising the predicted sales are about 6.97 thousand units.

The coefficient of 0.05546 for TV variable tells us that for every 1 thousand dollars we spend on TV advertising, sales increase by 0.055 thousand units.

The p value for TV variable is less than 2×10^{-16}, this tells us that the effect of TV advertising on sales is statistically significant.

The R squared value is 0.8122. This tells us that about 81.22 percent of the variation can be explained by TV advertising alone. This tells us that there is a very strong relationship between TV advertising and sales.

The F statistic is 856.2 and this has a p value of less than 2.2×10^{-16}, which indicates that the overall regression model is statistically significant.

Conclusion:

Our simple linear regression model indicates that TV advertising is a strong predictor of sales. Our sales increase with the increase in our TV advertising.

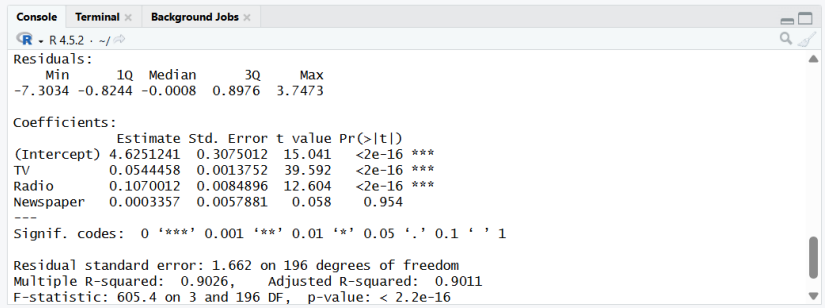
Task4(Multiple Regression Model)

Model specification

Sales = β0 + β1 × TV + β2 × Radio + β3 × Newspaper

From the R output the coefficients are:

Intercept = 4.6251
TV = 0.05445
Radio = 0.10700
Newspaper = 0.00034



Estimated regression equation

Sales = 4.6251 + 0.05445 × TV + 0.10700 × Radio + 0.00034 × Newspaper

Interpretation:

The value 4.625 is the predicted sales with TV, Radio & Newspaper values equal to 0.

The coefficient for TV advertising spends is 0.05445. That is, if TV advertising spend increases by 1 thousand dollars, the sales is about 0.054 thousand units, holding all the other variables constant.

The coefficient for Radio advertising spends is 0.10700. That is, if radio advertising increases by 1 thousand dollars, the sales are about 0.107 thousand units, holding all the other variables constant.

The coefficient for Newspaper advertising spend is 0.00034. This coefficient is very small in magnitude, indicating that there is almost no effect of newspaper advertising on sales in this model.

Statistical Significance:

The p value for TV advertising spend is less than 2×10^{-16}, indicating that there is a highly significant effect of TV advertising on sales.

Same story for the radio advertising effect with p value less than 2×10^{-16}.

The p value of newspaper advertising is 0.954 which is very large in relation to 0.05. This implies that the newspaper variable is not statistically significant.

The R squared is 0.9026 and therefore, about 90.26 percent of the variation in sales in this sample is accounted for by the three advertising variables in this sample.

The adjusted R squared is 0.9011 which is almost the same as the R squared and shows that this is a very good model.

The F statistic is 605.4, with p value less than 2.2×10^{-16} indicating that this regression model is statistically significant.

To provide a clearer comparison of the effects of Newspaper advertising, a simple regression model where Newspaper was the sole predictor was also fitted. The coefficient estimate for Newspaper in the simple regression model was 0.03832 with a p value of 0.0255, indicating a statistically significant small positive effect on sales. Nevertheless, in the multiple regression model the coefficient estimate for Newspaper was reduced to 0.00034 and the p value of 0.954, indicating that Newspaper was not statistically significant once the effects of TV and Radio advertising had been accounted for. As a result the apparent effect

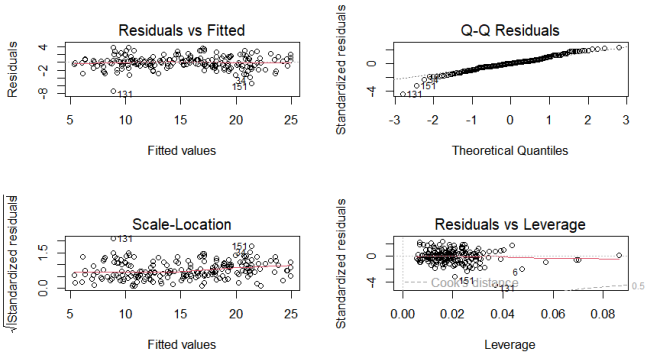
of Newspaper advertising in the simple model is due to its shared relationship with the other advertising channels.

Conclusion:

The multiple regression model indicates that TV advertising and Radio advertising spend both significantly affect sales. Newspaper advertising does not have a significant effect. More than 90 percent of the variation in sales is explained by the multiple regression model.

Task5

Residuals vs Fitted



This plot is used to assess the linearity assumption. The residuals seem to be randomly scattered around zero and there is no obvious pattern that forms. This implies that the linear relationship between the predictors and sales is suitable.

Normal Q-Q Plot

This plot is used as a visual test for the normality of the residuals. Most points are close to the straight reference line. This implies that the residuals are approximately normally distributed with only slight deviations at the ends.

Scale Location Plot

This plot is used to test the constant variance assumption. The spread of the residuals seems to be constant across the fitted values. The red line seems to be flat and thus the variance of the residuals are somewhat constant.

Residuals vs Leverage

This plot is used for identifying observations which are more influential than others. Most points lie on the left-hand side indicating low values of leverage and are still within the Cook's distance lines implying that we do not have highly influential observations.

Table (3) Flagged observations from regression diagnostics

Type of Flag	Observation Numbers	Criterion
Possible outliers	11, 34, 98, 131, 151, 155, 197	Absolute standardized residual > 2
Possible high leverage points	6, 17, 37, 76, 102, 129, 166	Leverage > 2p/n

The observations identified by the regression diagnostics are shown in Table 3. Several observations were flagged as possible outliers or high leverage points. These observations should be examined closely. Nevertheless, their presence does not necessarily indicate that the model is seriously mis-specified. In most respects, the diagnostic indicates that the multiple regression model is a reasonable fit. regression model is reasonably appropriate.

Conclusion:

The diagnostic plot shows that the main assumptions of regression are all quite well satisfied. The residuals seem to be randomly distributed, are approximately normal and the variance is constant. The model does not have high influential observations.

Task6

Model comparison

We compared the simple and multiple regression models using an analysis of variance test.

Model 1: without radio& newspaper, Contains only TV advertising as a predictor.

Model 2: has TV & Radio & Newspaper

The residual sum of squares for model 1 is = 1043.5 & for model 2 is = 541.2.

This substantial decrease in RSS indicates that inclusion of Radio and Newspaper improves the model's explanation of variation in sales.

The F statistic is 90.964 and p value is $< 2.2 \times 10^{-16}$. Therefore the transition from Model 1 to Model 2 is significantly better.

Table (4) Comparison of simple and multiple regression models

Measure	Model 1, TV only	Model 2, TV + Radio + Newspaper
R squared	0.8122	0.9026
Adjusted R squared	0.8112	0.9011
Residual standard error	2.2957	1.6617

Table 4 indicates that multiple regression model outperforms simple regression model. It explains a larger fraction of the variance in Sales and has smaller residual standard error. Therefore, the inclusion of Radio and Newspaper improves the overall fit of the model. The results of the multiple regression indicate that TV and Radio are statistically significant predictors of Sales while Newspaper is not statistically significant after other variables are controlled.

Table (5) Predicted effect of an additional 10 thousand dollars in advertising budget

Channel	Extra Budget	Predicted Increase in Sales
TV	10	0.544
Radio	10	1.070

Table 5 indicates that given an extension of the same amount to the budget, the increase in Sales predicted for Radio would be much greater than the increase in Sales predicted for TV. Thus, the allocation of the extra budget would favour Radio, provided that the regression relationship is linear and ceteris paribus.

Conclusion:

The multiple regression model significantly outperforms the simple regression model. The inclusion of other advertising channels better explains variation in sales. However, the regression results in Task 4 indicate that only TV and Radio advertising are statistically

significant predictors of sales. Newspaper advertising makes no significant contribution to the model.

Task7

Interpretation

First, sales observations were divided into two categories based on the mean of the sales observations. Observations with sales greater than the mean were classified as High sales and observations with sales less than the mean were classified as Low sales. 107 observations were in the high sales group, and 93 observations were in the low sales group.

A two-sample t test was subsequently conducted to determine if there is an effect of advertising expenditure between the High and Low sales groups.

TV advertising.

The average amount of TV advertising spent for the High sales group was 215.00 and the average amount spent for the Low sales group was 68.85. The p value for the t test was less than 2.2 X 10^-16 indicating that the two groups were very highly statistically different. consequently, increased TV advertising expenditure may be related to increased sales.

radio advertisements

The mean Radio advertising expenditure was 26.01 for the High sales group and 20.11 for the Low sales group. P value from the t test 0.0048 <0.05 so a significant difference exists between the two groups. So higher Radio advertising expenditure also seems to be correlated to higher sales.

Newspaper advertising expense.

The Average Newspaper advertising expense for High sales group was 33.04 and the Average Newspaper advertising expense for low sales group was 27.69. The t test value was .0817 larger than .05 Therefore it may be concluded that there are no statistically significant differences. Hence, Newspaper advertising spend is not highly correlated with sales.

The null hypothesis for the two-sample t-tests was that the mean advertising spend is the same in the High and Low Sales groups. The alternative hypothesis was that the mean advertising spend is different in the two groups. This test assumes that the observations are independent and that the observations in each group are approximately normally distributed. Because we used Welch's two-sample t-test are not assuming equal common variances.

Figure 9. TV advertising by High and Low Sales groups

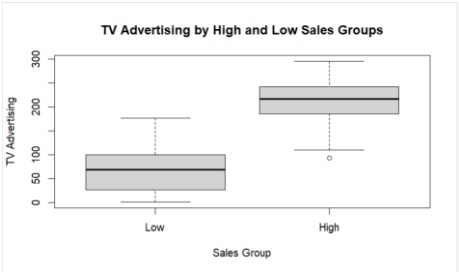


Figure 10. Radio advertising by High and Low Sales groups

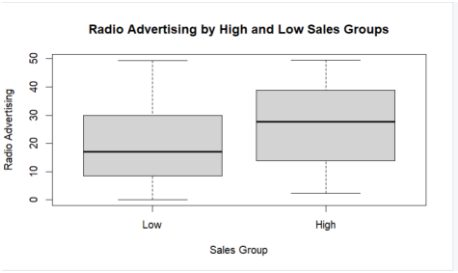
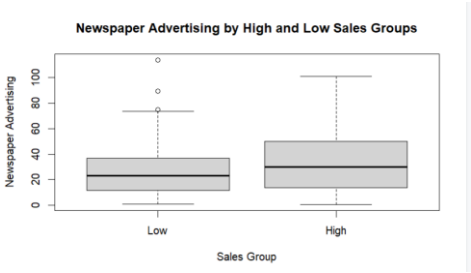


Figure 11. Newspaper advertising by High and Low Sales groups



The two-group split boxplots are consistent with the t-test results. TV advertising is considerably larger in the High Sales than the Low Sales group. Radio advertising is also somewhat larger in the High Sales group. Newspaper advertising does not show any large visual difference between the two groups, as consistent with the non-significant t-test.

Additional analyses using four Sales groups

To further investigates this relationship, sales were divided on quartiles to form four groups, labelled Low, Medium, High and Extreme. The group sizes were 52, 50, 48 and 50.

We next performed a one-way ANOVA to determine whether the mean value of advertising expenditure varied across these four Sales groups. Our null hypothesis for each test is that the mean value of advertising expenditure is the same across the four sales groups. Our alternative hypothesis is that at least one group mean is different.

ANOVA led to a statistically significant difference in mean TV ad across the four Sales groups, where p value < 2 × 10^-16. Radio ad also led to a statistically significant difference in the four groups, where p value = 3.84 × 10^-8. Newspaper ad led to no statistically significant difference, where p value = 0.245.

Next, I performed post hoc Tukey tests for Television and Radio advertising. For the Television, all pairwise group comparisons were statistically significant. Thus, Television advertising rises consistently as Sales increase from the Low to the Extreme group. For the Radio, only pairwise comparisons with the Extreme group were statistically significant. Thus, Radio advertising rises significantly only in the Extreme Sales group. We did not perform a post hoc test for Newspaper as the ANOVA for Newspaper was not significant.

Figure 12. TV advertising across Sales groups

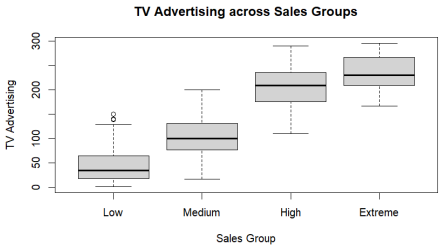


Figure 13. Radio advertising across Sales groups

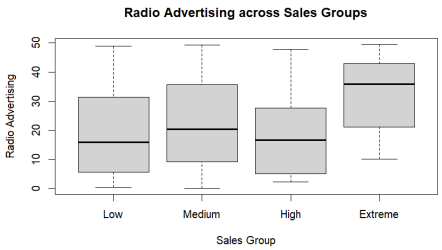
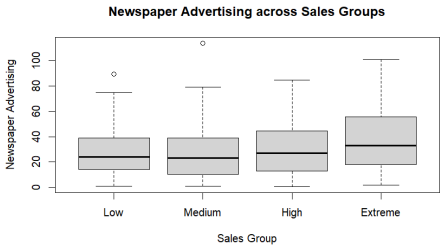


Figure 14. Newspaper advertising across Sales groups



The boxplots agree with the statistics. TV advertising shows a clear trend of increasing from Low through to the Extreme group, pointing to a strong relationship with Sales. Radio advertising is clearly higher on the Extreme group. This agrees with the ANOVA/Tukey results. Newspaper is not showing an obvious trend with the groups; this is consistent with the non-significant ANOVA result.

Conclusion

The conclusion is the same from the two-group t tests and the four-group ANOVA. TV advertising has the strongest relationship with Sales. Radio advertising also has a meaningful positive relationship with Sales. Newspaper advertising does not have a statistically significant relationship with Sales in the two-group t tests and the four-group ANOVA. Overall, it appears that TV and Radio advertising are a lot more important than Newspaper advertising in explaining the differences between sales.

Task8

Interpretation:

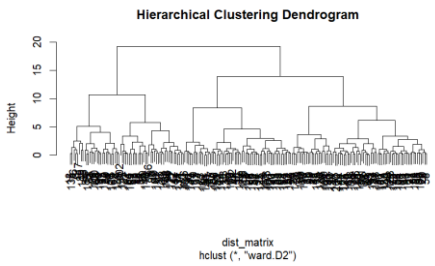
Principal Component Analysis was conducted on the standardised advertising variables TV, Radio, and Newspaper.

We have a solution with three principal components. PC1 accounts for 45.7% of the total variance in the advertising data, PC2 accounts for 32.77% of the total variance, and PC3 accounts for 21.53% of the total variance in the advertising data

The cumulative proportion shows that the first two principal components account for about 78.47% of the total variability in the data set; that is, most of the information contained in the advertising variables can be summarised by just using the first two principal components.

This result suggests that the advertising variables share some common patterns of variation and that it is possible to summarise most of the information with fewer components.

Hierarchical clustering was also conducted on the standardised advertising variables TV, Radio, and Newspaper. The dendrogram shows that observations can be clustered based on the similarity in their advertising expenditure patterns. This leads us to believe that there are patterns hidden in the dataset. In other words, the advertising data can be summarised by fewer common patterns determined through the PCA and Hierarchical Clustering.



Conclusion:

The PCA results indicate that the information in the variation of TV, Radio and Newspaper advertising can be collected into the first two principal components. This can also be supported by the hierarchical clustering results which indicate that observations can be clustered by their similar expenditure of advertising.

Task 9: Communicating the Results

Business Question:

The goal of this analysis is to find the advertising channel that most positively influences product sales to allow marketing managers to optimise the allocation of advertising budgets between TV, radio and newspaper advertising.

Summary of main findings:

Exploratory analysis revealed that budgets for TV advertising were the widest range of budgets in the dataset and budgets for radio and newspaper advertising were typically smaller. Initial understanding of the distribution of advertising spending and sales across observations was revealed using the histogram plots.

Correlation analysis found a strong positive relationship between Sales and TV advertising, a moderate positive relationship between Sales and Radio advertising, and a weak relationship between Sales and Newspaper advertising.

Regression analysis supported the results above. In the simple regression model, TV advertising alone is responsible for about 81 percent of the variability in sales. The addition of radio and newspaper advertising to the simple model increases the explanatory power of the model to about 90 percent. The results indicate that TV and radio advertising have statistically significant impacts on sales while newspaper advertising does not.

Regression model plots suggested that the assumptions of the regression model are well met. The residuals are random, approximately normal, and of constant variance. By analysis of variance, the testing indicated that the multiple regression model is significantly superior to the simple regression model.

Additional statistical tests comparing groups with high and low sales revealed statistically significant differences in levels of TV and radio advertising spend between the two groups, but not levels of newspaper spend.

Advertising allocation recommendations:

Statistical analysis shows that TV advertising should be given the largest share of the advertising budget, because it has the strongest overall relationship with sales and accounts for a large proportion of the variation in product demand.

Radio advertising should also be given a sizeable share of the budget, because it has a positive statistically significant effect on sales. Furthermore, the multiple regression model predicts that if an additional 10 thousand dollars were to be added to the Radio advertising budget, the increase in additional sales would be greater than if the same additional amount were added to the TV advertising budget.

Newspaper advertising appears to have little effect on sales in this data set. Therefore, marketing managers should assign a smaller share of the advertising budget to Newspaper advertising than to TV and Radio advertising.

Limitations:

The linearity assumption is made in this analysis between advertising spend and sales. Other potential drivers of sales such as product price, market competition, seasonal fluctuations in demand and promotional campaigns, have been excluded from the model thus the results should be considered as indicative of the relative importance of advertising channels rather than an exhaustive factor explaining sales.

References

Ashydv, n.d. Advertising dataset. Kaggle. Available at: https://www.kaggle.com/datasets/ashydv/advertising-dataset?resource=download [Accessed 12 March 2026].

R Core Team, 2024. R: A language and environment for statistical computing. Vienna: R Foundation for Statistical Computing.